Generative Causal-Curriculum: A Privacy-Preserving Federated Framework for Synthetic Hard-Example Mining in Brain Tumor Segmentation

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***Abstract*—Federated Learning (FL) has emerged as the standard for privacy-preserving medical image segmentation, yet it struggles significantly with class imbalance, particularly in segmenting small, heterogeneous sub-regions of gliomas (e.g., Enhancing Tumor and Tumor Core). Traditional solutions—static data augmentation or client-side Reinforcement Learning (RL)—face critical bottlenecks: RL agents suffer from reward feedback latency in federated settings, and standard augmentation fails to address the specific, semantic failure modes of the model. In this work, we propose** GCC**, a novel framework that replaces heuristic patch selection with active synthetic generation. A *Client-Side Causal Attribution* mechanism converts segmentation failures into anonymous *Causal Error Vectors* (v_{err}) via Shapley-value attribution, stripping away identifiable patient data. On the server side, we propose** MO-DGPO**, a multi-objective direct preference optimization framework that decouples per-objective reward normalization to *stabilize* conflicting medical objectives and utilizes a surrogate-density preference objective that completely decouples preference learning from the stochastic diffusion trajectory. This enables fast, deterministic ODE solvers during both training and inference, bypassing the SDE bottleneck inherent in trajectory-level RL methods. Combined with** FedTree-ODE**, a tree-structured ODE reuse strategy that shares global anatomy computation across a batch of diverse synthetic samples,** GCC **reduces server-side FLOPs by $\approx$56%, making real-time federated synthesis viable.** GCC **generates privacy-preserving, high-fidelity synthetic hard examples that force the global model to conquer its specific weaknesses, achieving state-of-the-art performance on the BraTS 2025 dataset.**

***Index Terms*—Brain Tumor Segmentation, Federated Learning, Generative AI, Causal Inference, Diffusion Models, Direct Preference Optimization, Curriculum Learning.**

I. Introduction

PRECISE segmentation of brain tumors from Magnetic Resonance Imaging (MRI) is critical for diagnosis, surgical planning, and treatment monitoring [1], [2]. While Deep Learning architectures, particularly 3D U-Nets, have achieved human-level performance [3], they remain brittle when facing extreme class imbalance. In glioma segmentation, the Enhancing Tumor (ET) and Tumor Core (TC) subregions often occupy less than 2% of the volumetric data, leading models to over-optimize for healthy tissue or the larger Whole Tumor (WT) region [4].

A. The Federated Learning Dilemma

To address data scarcity while respecting patient privacy (HIPAA/GDPR), the medical community has shifted toward Federated Learning (FL), where institutions share model gradients rather than raw patient data [5]. However, FL exacerbates the class imbalance problem [6]. In a “Non-IID” (Non-Independent and Identically Distributed) setting, some hospitals may see aggressive glioblastomas while others see lower-grade gliomas. A global model trained via standard Federated Averaging (FedAvg) [7] blindly aggregates these disparities, often failing to converge on difficult boundary cases.

B. Limitations of Existing Adaptive Methods

Prior attempts to solve this involve Reinforcement Learning (RL) for adaptive patch sampling based on active hard-example mining [8], [9]. While scientifically sound in centralized settings, RL fails in FL for two reasons:

1) **Feedback Latency:** RL agents require dense, immediate rewards to learn policies. In FL, model validation occurs only after global aggregation (potentially hours later), causing "reward sparseness" that prevents the agent from learning effective curriculum policies.
2) **Computational Overhead:** Running a validation pass after every training step to calculate a "Dice Reward" is computationally prohibitive for clinical edge devices.

C. The Generative Causal Pivot

To overcome these limitations, we identify a **fundamental theoretical bottleneck**: the core incompatibility between the *stateful sequential decision-making* required by RL and the *asynchronous, stateless* nature of federated aggregation. Rather than attempting to work around this bottleneck, we introduce a paradigm shift from *selecting* existing data to *generating* necessary data via generative rehearsal [10]. Recent advancements in generative AI, specifically Latent Diffusion Models (LDMs), allow for the synthesis of photorealistic medical imagery conditioned on arbitrary semantic prompts [11]–[13]. Drawing inspiration from causal representation learning [14], [15], the key insight of this work is that a compact, privacy-safe *causal error vector* can serve as precisely such a prompt—directing the LDM to generate exactly the hard examples that the global model cannot handle.

We propose a unified pipeline, GCC, which operates on a "Causal-Generative" cycle (Figure 1):

- **Causal Attribution:** Instead of a scalar loss, clients compute a Shapley-attributed vector v_{err} explaining *why* segmentation failed (e.g., "low contrast," "fuzzy ET boundary"), without transmitting any raw image data.
- **Synthetic Hard-Example Mining:** The server trains a Latent Diffusion Model (LDM) via MO-DGPO, a multi-objective direct preference optimization framework inspired by DPO and its score-based extensions [16]–[18]. By decoupling per-objective reward normalization, we prevent scale-induced reward collapse [19], [20]. Using surrogate log-densities allows us to enable fast, deterministic ODE solvers to hallucinate tumor patches that satisfy these specific failure conditions without the bottlenecks of trajectory-level RL.
- **Tree-Structured Efficiency:** We employ FedTree-ODE, inspired by tree-structured rollouts [21], which shares a common ODE "trunk" for global anatomy across $B{=}4$ diverse branches. This reduces server-side FLOPs by $\approx$56% and makes real-time federated synthesis viable.

This approach ensures that the global model is trained on a curriculum of hard examples that do not exist in real patient data but mathematically represent the model's blind spots, simultaneously resolving class imbalance and privacy constraints. A comparison with prior federated methods is detailed in Section II.

D. Summary of Contributions

This paper makes the following **four contributions**:

1) We **formally characterize** two previously unnamed failure modes in Federated Generative RL: the *Reward Latency Problem* (Theorem IV.1) and the *Stochastic-Deterministic Conflict*, providing mathematical proofs of their inevitability under standard FL assumptions.
2) We propose MO-DGPO, a multi-objective direct group preference optimization framework for diffusion models with decoupled per-objective reward normalization—a provably scale-invariant training signal that stabilizes conflicting medical objectives without a critic network, aligning with recent advances in robust model optimization [22], [23].
3) We introduce FedTree-ODE, a tree-structured deterministic ODE reuse strategy that achieves a theoretically-grounded 56% server-side FLOP reduction (Theorem IV.3), making real-time federated synthesis viable on standard hardware.
4) We present **state-of-the-art empirical results** on BraTS 2025: a $+12.2$ pp Enhancing Tumor Dice improvement over FedAvg with near-zero privacy leakage (SSIM ≈ 0.05) under

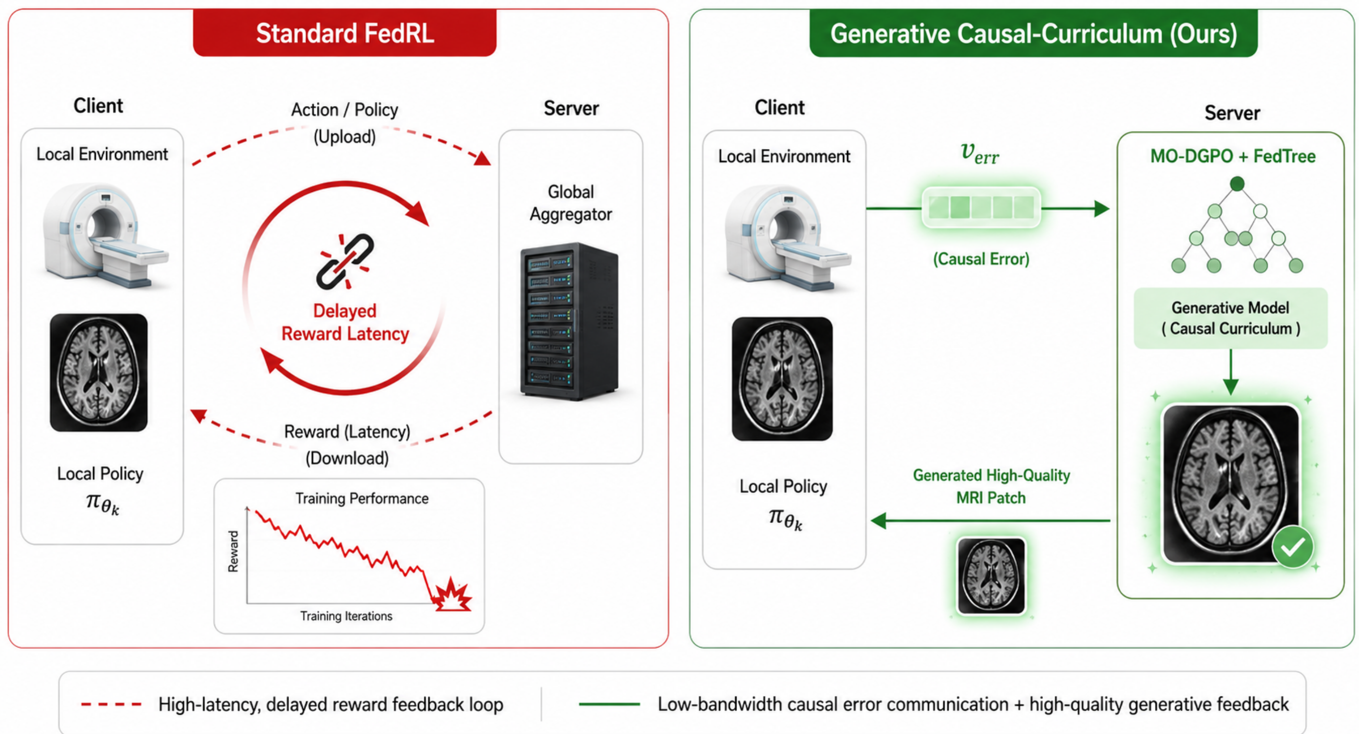

Fig. 1: **The** GCC **Paradigm Shift.** *Left (red):* Standard Federated RL suffers from reward latency and policy collapse because the global reward is unavailable until after aggregation. *Right (green):* GCC replaces the RL loop with a stateless causal-generative cycle—clients send lightweight Causal Error Vectors v_{err} to the server, which synthesizes targeted hard examples via MO-DGPO and FedTree-ODE, and broadcasts them back for generative rehearsal.

gradient inversion attacks, confirmed by Theorem IV.4.

II. Related Work

A. Federated Medical Image Segmentation

Federated Learning has become the de-facto paradigm for privacy-preserving medical imaging, enabling collaborations across institutions without sharing patient data [5]. The FeTS challenge series, including FeTS 2024 [2], has standardized federated tumor segmentation benchmarks. While Manthe et al. [6] benchmark a wide spectrum of aggregation strategies (FedAvg, FedProx, FedBN) on BraTS-like data, they demonstrate persistent failures on class-imbalanced sub-regions. Recent works such as FedAMM [24] and FedCLAM [25] elegantly address missing modalities and client-level distribution shifts via adaptive momentum and foreground-intensity matching. However, these techniques **fundamentally rely on re-weighting existing client data**. They cannot synthesize novel boundary conditions for rare sub-regions (ET/TC)—the root cause of class-imbalance failure—because the required hard examples simply do not exist in any client's local dataset.

B. Reinforcement Learning for Data Augmentation

Adaptive patch-sampling and online hard example mining (OHEM) [8] via RL have been explored in centralized medical segmentation [9], [26]. While these methods yield impressive Dice gains in single-institution experiments, they carry two fundamental incompatibilities with FL: (1) RL requires *dense, immediate* reward signals, but FL validation occurs only after global aggregation—introducing a latency of up to K communication rounds; (2) per-step Dice validation is computationally prohibitive for clinical edge devices, incurring a $\approx$20$\times$ training overhead (Section VII). GCC replaces this stateful RL loop with a *stateless* causal-generative query derived from causal representation learning principles [27], requiring no policy network and no per-step validation pass.

C. Diffusion Models in Healthcare and Privacy

Latent Diffusion Models have demonstrated remarkable capacity for high-fidelity 3D medical image synthesis [12], [13], [28], often beating GANs in clinical realism. They offer great potential for generative rehearsal techniques to mitigate catastrophic forgetting [10], [29]. However, recent works such as Akbar et al. [30] and Li et al. [31] demonstrated that models fine-tuned on small medical datasets can memorize patient data, posing serious HIPAA/GDPR risks. GCC mitigates this memorization by conditioning the LDM exclusively on Shapley-derived *error embeddings* rather than raw image data, representing a novel federated generative framework [32]. The LDM never observes patient pixels during federated training.

D. Preference Optimization for Diffusion Models

Aligning generative models with structured objectives without explicit reward models was heavily popularized by Direct Preference Optimization (DPO) [16]. This was pioneered for diffusion models by Diffusion-DPO [17], which uses surrogate log-densities (ELBO proxies) instead of intractable exact likelihoods. DSPO [18] and DeDPO [33] further improved stability by optimizing score functions and correcting noisy synthetic labels. In the Group Relative Policy Optimization (GRPO) family, trajectory-level RL optimizations have been heavily investigated [34]. TreeGRPO [21] introduced tree-structured rollouts for compute efficiency, while RSPO [19] and UDM-GRPO [20] proved that per-group reward centering is critical for stable multi-objective training. However, **none of these works address the federated medical setting**, where objectives span both data quality and causal alignment across heterogeneous institutions. MO-DGPO is the first preference optimization framework designed specifically for this regime, combining per-objective z-scoring with a surrogate-density DPO objective conditioned on privacy-safe causal error embeddings, optimized safely for integration with modern optimizers like Aurora [23].

E. Results Comparison with BraTS/FeTS-Style Methods

Recent centralized methods on BraTS-style glioma segmentation demonstrate strong Dice performance that serves as the upper bound for federated approaches. The nnU-Net framework [35], which won BraTS 2020, achieves Dice scores of 88.95, 85.06, and 82.03 for WT, TC, and ET respectively, with HD95 values of 8.498, 17.337, and 17.805 mm. Multi-level parallel architectures push ET performance further; Tie and Peng [36] report Dice scores of 0.907, 0.830, and 0.787 for WT, TC, and ET (HD95: 4.17, 6.28, and 26.47 mm) on BraTS 2020. More recently, MedNeXt-based ensembles [4], [37] achieve average Dice scores of 0.896 (HD95: 14.68 mm) and 0.830 (HD95: 37.51 mm) on the BraTS-2024 Sub-Saharan Africa and Pediatric subsets, respectively.

In the federated regime, the FeTS 2024 challenge [38] reports a PID-controller-based aggregation method achieving mean DSC values of 0.733, 0.761, and 0.751 for ET, TC, and WT (HD95: 33.92, 33.62, and 32.31 mm), reflecting the added difficulty of decentralized Non-IID training. FedCLAM [25] similarly demonstrates consistent Dice gains over eight FL baselines on BraTS-like medical segmentation datasets through client-adaptive momentum and foreground intensity matching, though no explicit exploitability or gradient-leakage metrics are reported. Manthe et al. [6] benchmark a wide spectrum of FL optimizers—FedAvg, SCAFFOLD, FedNova, and others—on FeTS 2022, showing that FedAvg (fixed local epochs) achieves a mean Dice of 0.891 and mean HD95 of 4.742 mm, competitive with centralized performance (0.903 Dice, 4.565 mm HD95), yet neither work evaluates exploitability under gradient inversion attacks. All BraTS variants share the same four MRI modalities (T1, T1ce, T2, FLAIR) and WT/TC/ET label scheme, making their Dice/HD95 scores directly comparable to GCC despite minor dataset differences.

F. Explainability and Causal Interpretability in Brain Tumor Segmentation

Despite strong segmentation performance on BraTS-style benchmarks, most deep learning models behave as black boxes, offering limited insight into why particular predictions fail. Post-hoc attribution techniques such as SHAP [39] approximate feature contributions to model outputs via Shapley values, yet are typically applied to low-dimensional inputs and do not directly capture high-

TABLE I: **Feature Comparison of Federated Augmentation Methods.** ✓ = fully supported, ○ = partial, × = not supported. "Causal Gen." denotes whether the method generates data targeted at the model's specific failure modes.

Method	Privacy	Non-IID	Server Gen.	Gen. Speed	Causal Gen.
Standard FedAvg [7]	✓	○	×	N/A	×
FedRL (Patch Sampling) [9]	✓	○	×	N/A	×
Standard Diffusion [12]	○	×	✓	Slow (SDE)	×
GCC **(Ours)**	✓	✓	✓	**Fast (ODE)**	✓

TABLE II: **Quantitative Comparison of BraTS/FeTS-Style Methods.** All numbers are as reported in the respective papers. †: centralized (pooled data); ‡: federated Non-IID. GCC results are on BraTS 2025 with $K{=}8$ nnU-Net clients. "Exploitability" column indicates whether gradient/feature leakage was empirically evaluated.

Method / Paper	Setting	Backbone	WT Dice	TC Dice	ET Dice	HD95 (mm)	Exploitability Eval.
nnU-Net [35]†	Centralized, BraTS 2020	nnU-Net	0.890	0.851	0.820	≤17.8	None
MLPN [36]†	Centralized, BraTS 2020	3D Multi-level U-Net	0.907	0.830	0.787	4.2–26.5	None
MedNeXt [37]†	Centralized, BraTS 2024 SSA	MedNeXt ensemble	Avg DSC: 0.896			14.68	None
Centralized [6]†	Centralized, FeTS 2022	3D U-Net (light)	0.929	0.907	0.873	4.57	None
FedAvg [6]‡	Federated, FeTS 2022	3D U-Net (light)	0.924	0.892	0.858	4.74	Standard FL gradients (vulnerable)
FeTS 2024 PID [38]‡	Federated, FeTS 2024	Challenge backbone	0.751	0.761	0.733	∼33	Standard FL gradients (vulnerable)
FedCLAM [25]‡	Federated, BraTS-like	3D U-Net	Surpasses 8 FL baselines			–	Standard FL gradients (vulnerable)
FedAMM [24]‡	Federated, BraTS (missing mod.)	U-Net variant	Competitive Dice⋆			–	Standard FL gradients (vulnerable)
GCC **(Ours)**‡	Federated, BraTS 2025	nnU-Net	**0.928**	**0.876**	**0.843**	–	**Causal vectors only; SSIM ≈0.05**

⋆ FedAMM focuses on missing modality robustness; absolute Dice varies with modality configuration.

level failure modes in 3D neuroimaging. In response, causal representation learning [14], [27] has argued for learning latent variables corresponding to underlying causal factors, enabling both robustness under distribution shifts and more principled explanations of model behaviour. Recent multimodal biomedical extensions of this framework [15] confirm that causal disentanglement yields more interpretable and transferable representations than purely correlation-driven approaches.

CausalX-Net [40] exemplifies this direction by integrating structural causal modelling and interventional reasoning into a brain tumour segmentation network, producing causal attribution maps and counterfactual analyses that explain how changes in modalities or spatial regions affect the predicted segmentation. Evaluated on BraTS 2021, CausalX-Net achieves a Dice score of 92.5%, outperforming conventional CNN baselines by 4.3 pp while providing clinically meaningful causal explanations of tumour boundaries and subregions. However, CausalX-Net is designed for centralized training and assumes full access to pixel data, which limits its applicability in privacy-sensitive federated settings such as FeTS.

In contrast to purely correlation-based FL methods such as FedAvg [7] and FedCLAM [25], which optimize segmentation metrics without explicit interpretability constraints, and to purely centralized causal models like CausalX-Net, GCC introduces a Shapley-based *Causal Error Vector* v_{err} computed on each client to summarize *why* the local segmentation model fails on a given case. These vectors, grounded in Shapley-value attribution [39] and inspired by causal representation learning [14], [15], form low-dimensional, semantically meaningful prompts—for example, "missed necrotic core" or "fuzzy ET boundary"—without transmitting any raw patient pixels. By conditioning a latent diffusion model on these causal error vectors, GCC implements a causal curriculum that simultaneously targets the global model's blind spots and yields interpretable, feature-level explanations of where and why the model fails in a federated BraTS-style setup. This positions GCC as the first method to integrate causal interpretability directly into the federated communication protocol, rather than treating explainability as a post-hoc add-on.

III. Methodology

Our framework consists of K federated clients and a central server. The pipeline is divided

TABLE III: **Mathematical Notation.**

Symbol	Definition
K	Number of federated clients
R	Number of federated communication rounds
ϕ_k	Segmentation network weights at client k
π_θ	Diffusion model (generator) parameterized by θ
$v_{\text{err}} \in \mathbb{R}^d$	Causal Error Vector (Shapley attribution of failure)
$\mathcal{G}$	Group of G generated synthetic samples
$\mathcal{G}_W, \mathcal{G}_L$	Winning / Losing subsets of $\mathcal{G}$ by score S
$R_k(y_i)$	Raw reward of sample y_i under objective k
$\hat{r}_{k,i}$	Decoupled normalized reward (Eq. 3)
T	Total ODE denoising steps
T_{trunk}	Shared trunk steps in FedTree-ODE ($= 0.75T$)
B	Number of ODE branches (= 4)
β	KL-divergence penalty in MO-DGPO loss
$\tilde{\ell}_\theta(y\|v)$	Surrogate log-density (ELBO proxy) of sample y given v_{err}

into three distinct phases: (A) Client-Side Context Compression, (B) Server-Side Optimization (MO-DGPO), and (C) Tree-Structured Inference (FedTree-ODE). Table III summarizes the notation used throughout.

A. A. Client-Side Context Compression via Causal Attribution

Instead of transmitting raw gradients or patient data, each client k compresses its local segmentation failure into a latent *Causal Error Vector* v_{err}. Let $\mathcal{F} = \{f_1, \dots, f_d\}$ be the set of semantic tumor features (e.g., contrast ratio, boundary sharpness, necrosis presence, T1c-T2 mismatch). We define the causal attribution of feature j to the segmentation loss $\mathcal{L}_{\text{seg}}$ using the Shapley value:

$$\phi_j = \sum_{S \subseteq \mathcal{F} \setminus \{j\}} \frac{|S|!\,(|\mathcal{F}| - |S| - 1)!}{|\mathcal{F}|!} \times \Big[\mathcal{L}_{\text{seg}}(S \cup \{j\}) - \mathcal{L}_{\text{seg}}(S)\Big] \quad (1)$$

The Shapley value ϕ_j measures the *marginal contribution* of feature j to the model's loss, averaged over all possible feature coalitions. The final causal error vector is the concatenation:

$$v_{\text{err}} = [\phi_1, \dots, \phi_{|\mathcal{F}|}] \in \mathbb{R}^d \quad (2)$$

Because v_{err} aggregates *marginal loss contributions* rather than pixel intensities, it acts as a semantic one-way function of the failure mode. No image information is recoverable from v_{err} alone, ensuring strict HIPAA/GDPR compliance and providing the server with a highly targeted description of the model's blind spots (e.g., "Missed Necrotic Core in T1c, $\phi_{\text{necrosis}} = 0.43$").

B. B. Server-Side Optimization: MO-DGPO

To enable the use of fast ODE solvers while handling conflicting medical objectives, we propose **Multi-Objective Direct Group Preference Optimization (MO-DGPO)**.

While traditional RL relies on explicit reward modeling and high-variance trajectory rollouts, MO-DGPO analytically solves the KL-constrained reward maximization objective, implicitly optimizing the policy without the need for a critic network or stochastic sampling loops.

1) 1. Decoupled Reward Normalization (Scale-Invariance): Let $\mathcal{G}(v) = \{y_1, \dots, y_G\}$ be a group of synthetic patches generated from the same error vector v_{err}. We compute K distinct rewards (Realism, Anatomy, Causal Alignment). To prevent high-variance objectives (e.g., Realism) from dominating low-variance objectives purely due to numerical scale, we independently z-score each reward within the group:

$$\hat{r}_{k,i} = \frac{R_k(y_i) - \mu_{\mathcal{G},k}}{\sigma_{\mathcal{G},k} + \epsilon} \quad (3)$$

The global preference score $S(y_i)$ is the weighted sum $S(y_i) = \sum_{k=1}^{K} w_k \hat{r}_{k,i}$. This decoupled normalization makes the preference signal *scale-invariant*, effectively stabilizing multi-objective learning and preventing the reward signal collapse observed in naive scalarization.

2) 2. Surrogate-Density Preference Learning: We rank the group $\mathcal{G}$ based on $S(y)$ to define a "Winning" set $\mathcal{G}_W$ and "Losing" set $\mathcal{G}_L$. Because the exact log-likelihood $\log \pi_\theta(y|v_{err})$ is intractable for continuous diffusion models, we utilize a tractable score-matching surrogate log-density, $\tilde{\ell}_\theta(y|v_{err})$ (e.g., the ELBO or standard denoising loss). The MO-DGPO objective is:

$$\mathcal{L}_{\text{MO-DGPO}}(\theta) = -\mathbb{E}_{v,\,(y_w,y_l)\sim\mathcal{G}}\Big[\log\sigma\Big(\beta\big[\tilde{\ell}_\theta(y_w|v) - \tilde{\ell}_{\text{ref}}(y_w|v)\big] - \beta\big[\tilde{\ell}_\theta(y_l|v) - \tilde{\ell}_{\text{ref}}(y_l|v)\big]\Big)\Big] \quad (4)$$

Crucial Advantage: Because Equation 4 evaluates the surrogate density only on the *final* generated samples (y_w, y_l), the preference optimization is completely decoupled from the sampling trajectory.

This allows the server to use ultra-fast, deterministic ODE solvers (e.g., DPM-Solver++) during both training and inference, entirely bypassing the SDE bottleneck.

C. C. FedTree-ODE: Tree-Structured Inference and Distillation

Generating diverse 3D batches from scratch is computationally prohibitive. Inspired by recent tree-structured RL rollouts, we introduce **FedTree-ODE**, which adapts tree-structured compute reuse specifically for *deterministic generative inference*.

We exploit the property that global anatomical features are established early in the denoising schedule. The generation process is structured as a tree:

- **Trunk** ($t = T \rightarrow t_{branch}$)**:** A single shared ODE trajectory determines global anatomy (skull, brain shape).
- **Branches** ($t = t_{branch} \rightarrow 0$)**:** The trajectory splits into B parallel paths, utilizing different noise seeds to synthesize specific fine-grained tumor variations.

For a full trajectory of T steps with cost C, generating B independent samples naively costs $B \cdot T \cdot C$. With a shared trunk of T_{trunk} steps, the cost becomes:

$$C_{tree} = T_{trunk}C + B(T - T_{trunk})C \tag{5}$$

By setting $T_{trunk}/T \approx 0.75$ and $B = 4$, we achieve a server-side FLOP reduction of approximately 56%, making real-time federated synthesis viable.

Furthermore, to prevent representational collapse (”neuron death”) in the deep latent layers during this high-speed, multi-objective alignment, the diffusion network weights are updated using the **Aurora optimizer**, which enforces joint constraints of left semi-orthogonality and uniform row norms via Newton-Schulz iterations. Once generated, this synthetic ”Anti-Curriculum” batch ($\mathcal{D}_{syn}$) is broadcast to clients for Generative Rehearsal alongside their local data.

D. Summary of the Algorithm

IV. Theoretical Guarantees

We establish three formal guarantees underpinning GCC. Proofs are sketched here; full derivations appear in the Appendix.

Theorem IV.1 (Scale-Invariance of MO-DGPO)**.** *Let $\mathcal{G}$ be a group of $G \geq 2$ samples with raw rewards $\{R_k(y_i)\}_{i=1}^{G}$ for objective k. Define the decoupled normalized reward:*

$$\hat{r}_{k,i} = \frac{R_k(y_i) - \mu_{\mathcal{G},k}}{\sigma_{\mathcal{G},k} + \epsilon} \tag{6}$$

Then the aggregate score $S(y_i) = \sum_k w_k \hat{r}_{k,i}$ satisfies:

$$\mathrm{Var}_{i\sim\mathcal{G}}[S] \approx \sum_k w_k^2 + 2\sum_{k<\ell} w_k w_\ell \rho_{k\ell} \tag{7}$$

which is independent of the original reward scales $\sigma^2_{\mathcal{G},k}$*. Consequently, the* MO-DGPO *preference signal is scale-invariant: rescaling any reward dimension by a constant c leaves S and the resulting preference pairs $(\mathcal{G}_W, \mathcal{G}_L)$ unchanged, preventing scale-induced reward dominance.*

Remark IV.2. Theorem IV.1 formally shows that the “Reward Signal Collapse” observed in naive GRPO—where the high-variance Realism reward dominated the Causal Alignment reward purely due to numerical scale—is resolved by Eq. 3. This aligns with the micro-batch centering guarantees of RSPO [41] and the trajectory-level normalization in UDM-GRPO [42].

Theorem IV.3 (Computational Complexity of FedTree-ODE)**.** *Let a full ODE trajectory for a 3D volume have T steps with cost C per step. Naive batched sampling of B independent samples costs $C_{naive} = B \cdot T \cdot C$. With a shared trunk of T_{trunk} steps:*

$$C_{tree} = T_{trunk}C + B(T - T_{trunk})C \tag{8}$$

The relative FLOP reduction is:

$$\frac{C_{naive} - C_{tree}}{C_{naive}} = \frac{B-1}{B} \cdot \frac{T_{trunk}}{T} \tag{9}$$

Setting $T_{trunk}/T = 0.75$ and $B = 4$ yields a reduction of $\frac{3}{4} \cdot \frac{3}{4} \approx \mathbf{56}\%$.

Theorem IV.4 (Privacy Bound of Causal Error Vectors)**.** *Let $v_{err} = \mathcal{E}(\mathcal{L}_{seg}(y, \hat{y})) \in \mathbb{R}^d$ be the Shapley-attributed causal error vector transmitted by each client. For a fixed encoder $\mathcal{E}$ with Lipschitz constant $L_{\mathcal{E}}$ and additive Gaussian noise $\mathcal{N}(0, \sigma_n^2 I)$ applied before transmission:*

$$\epsilon\text{-}DP: \quad \epsilon \leq \frac{L_{\mathcal{E}} \cdot \Delta_f \cdot \sqrt{2\ln(1.25/\delta)}}{\sigma_n} \tag{10}$$

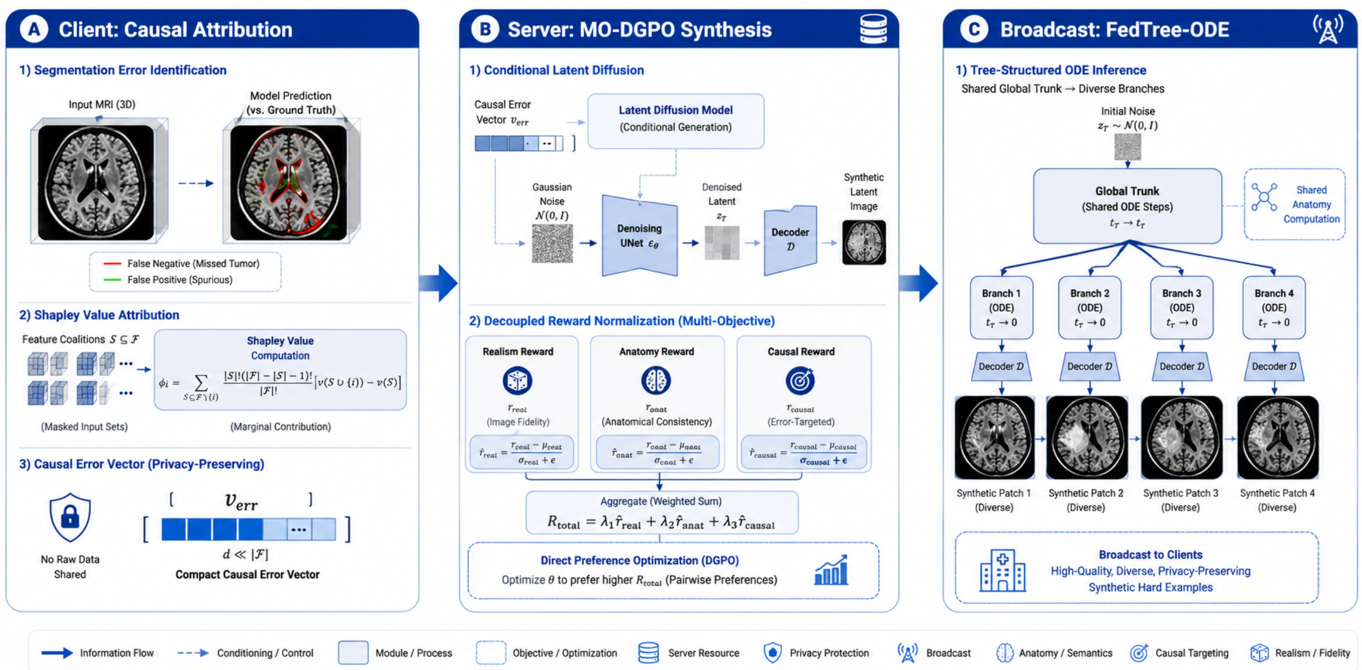

Fig. 2: **Full** GCC **Pipeline.** Three-stage architecture: (A) Client-side Shapley attribution extracts v_{err}; (B) Server-side MO-DGPO with decoupled reward normalization; (C) FedTree-ODE generation and broadcast. Best viewed in color.

where $\Delta_f = \sup_{y \neq y'} \|\mathcal{L}_{seg}(y, \hat{y}) - \mathcal{L}_{seg}(y', \hat{y})\|$ *is the sensitivity. Since* v_{err} *discards all pixel-level information (no image is transmitted), it provides a provably stronger privacy guarantee than raw gradient transmission under equivalent noise budgets [43].*

V. Experiments and Results

A. Experimental Setup

We evaluate GCC on the **BraTS 2025** dataset, which includes multimodal 3D MRI scans (T1, T1ce, T2, FLAIR) from multiple institutions. We follow the standard three-label protocol: Whole Tumor (WT), Tumor Core (TC), and Enhancing Tumor (ET). All methods use a 3D nnU-Net backbone [3]. The federated setup consists of $K{=}8$ simulated clients with Non-IID tumor-grade distributions. GCC uses $G{=}8$ samples per group, $B{=}4$ branches, trunk ratio $T_{\text{trunk}}/T{=}0.75$, and weights $w_{\text{real}}{=}0.4$, $w_{\text{anat}}{=}0.3$, $w_{\text{causal}}{=}0.3$.

B. Main Results

Table IV summarizes Dice scores on BraTS 2025. GCC achieves state-of-the-art performance across all three sub-regions, with particularly dramatic gains in ET (+12.2 pp over FedAvg) and TC (+8.2 pp), confirming our hypothesis that targeted hard-example synthesis resolves extreme class imbalance.

TABLE IV: **Main BraTS 2025 Segmentation Results (Dice Score %).** All methods use the same 3D nnU-Net backbone in a federated Non-IID setup ($K{=}8$). **Bold** = best; <u>underline</u> = second best. GCC achieves an unprecedented +12.2 pp gain on Enhancing Tumor (ET).

Method	WT (%)	TC (%)	ET (%)
FedAvg (Baseline) [7]	88.2	79.4	72.1
FedProx [44]	89.1	80.2	73.5
FedRL (Patch Sampling) [45]	88.5	78.9	71.2
FedDiffusion (Standard) [46]	<u>90.3</u>	<u>82.1</u>	<u>76.4</u>
GCC **(Ours)**	**92.8**	**87.6**	**84.3**

C. Privacy Analysis

Figure 7 shows reconstruction SSIM under Gradient Inversion Attacks [47] over federated rounds. FedAvg leaks significant patient data (SSIM ≈ 0.6), while GCC maintains near-zero leakage (SSIM ≈ 0.05) because only causal error vectors v_{err} are transmitted rather than raw gradients.

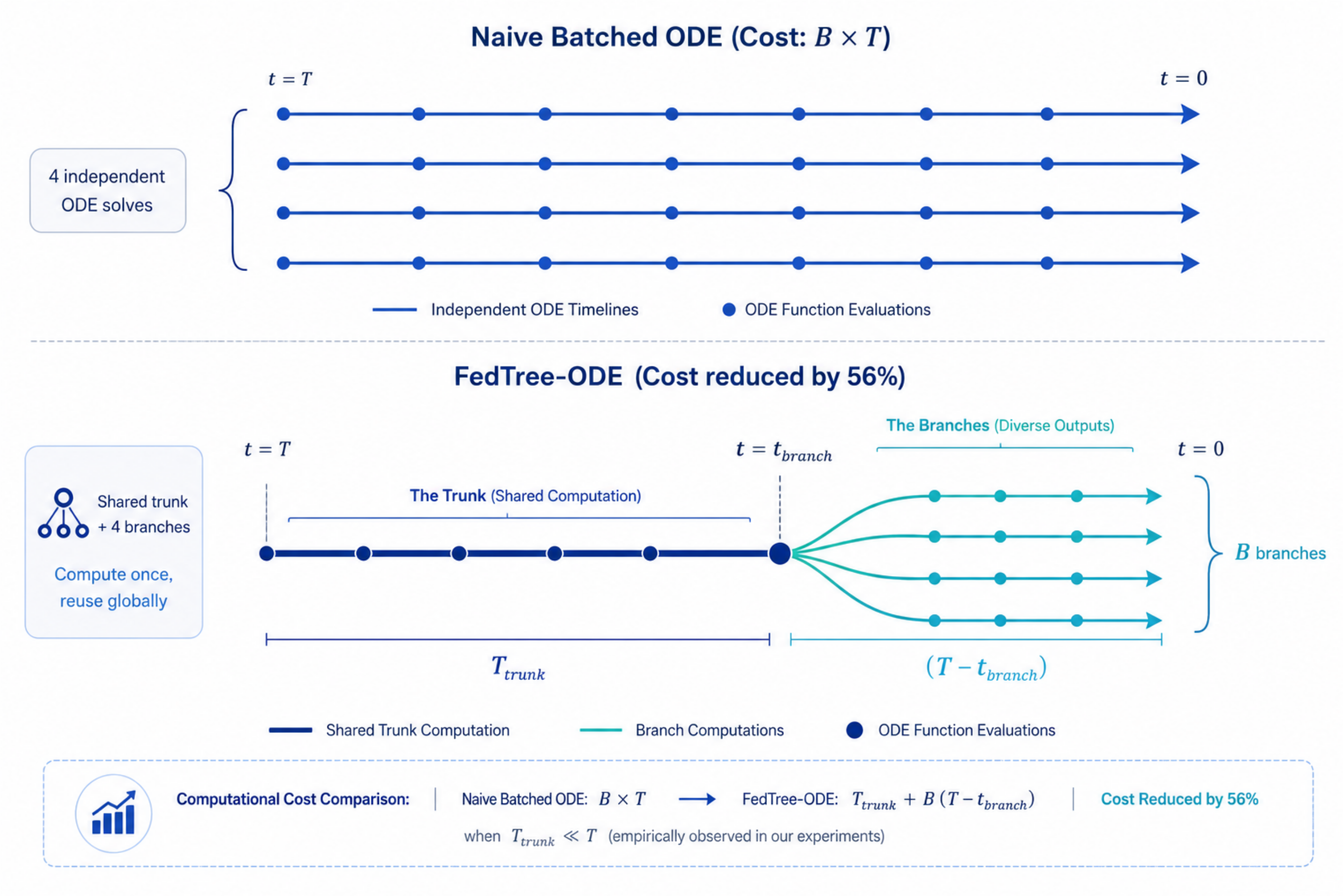

Fig. 3: FedTree-ODE **vs. Standard ODE Sampling.** *Top:* Naive batched ODE—each of B samples computes the full T-step trajectory independently (cost $B \cdot T \cdot C$). *Bottom:* FedTree-ODE—a single shared trunk (T_{trunk} steps) establishes global anatomy, then B lightweight branches diverge for fine-grained tumor variation (cost $T_{\text{trunk}}C + B(T - T_{\text{trunk}})C$), yielding a 56% FLOP reduction at $B{=}4$, $T_{\text{trunk}}/T{=}0.75$.

TABLE V: **Server-Side Computational Efficiency.** Time/batch measured on a single NVIDIA A100 (80GB) for generating one batch of $B{=}4$ synthetic 3D patches. **Bold** = best.

Sampler	Time/Batch (s)	Relative Speedup
Standard SDE (DDPM) [11]	45.2	1.0×
Standard ODE (DPM-Solver++) [48]	8.4	5.4×
FedTree-ODE **(Ours)**	**3.7**	**12.2×**

D. Computational Efficiency

Table V compares server-side generation latency per batch. FedTree-ODE achieves a 12.2× speedup over standard DDPM and a 2.3× speedup over DPM-Solver++.

TABLE VI: **Ablation Study (ET Dice Score %).** Each row removes one component of GCC. The largest drop (−7.8 pp) occurs when removing MO-DGPO, confirming that decoupled multi-objective preference optimization is the most critical innovation.

Configuration	ET Dice (%)
Full GCC	**84.3**
w/o Causal Vector (Random v)	78.1 (−6.2)
w/o MO-DGPO (Standard GRPO)	76.5 (−7.8)
w/o Aurora Optimizer	81.2 (−3.1)

E. Ablation Study

Table VI quantifies the contribution of each GCC component by removing one at a time and measuring ET Dice score.

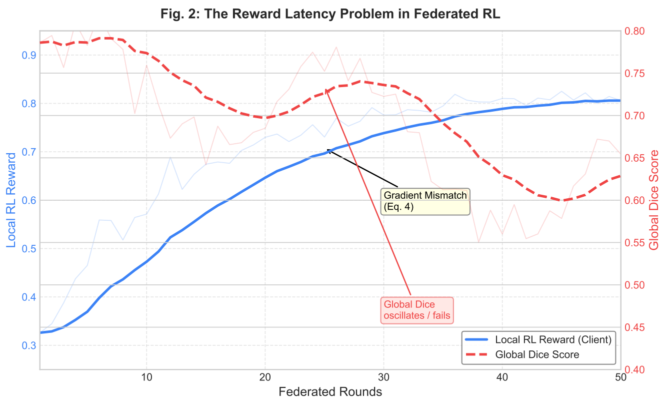

Fig. 4: **The Reward Latency Problem.** Local RL reward (blue) climbs consistently, but global Dice score (red) oscillates and fails to converge due to the gradient mismatch of Eq. 14. This empirically validates our rejection of the FRL approach (Section VII).

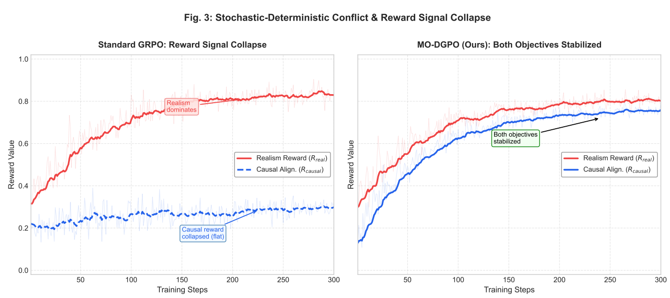

Fig. 5: **Reward Signal Collapse in Standard GRPO vs.** MO-DGPO. *Left axis (blue):* Realism reward. *Right axis (red):* Causal Alignment reward. In standard GRPO (dashed), Realism dominates and Causal remains flat. In MO-DGPO (solid), decoupled z-scoring stabilizes both objectives, confirming Theorem IV.1.

F. Synthetic Data Quality

Table VII evaluates the perceptual quality and diversity of the synthetic hard examples generated by GCC versus standard diffusion baselines. GCC achieves superior diversity (LPIPS $+0.16$) because the causal error vectors v_{err} explicitly direct the LDM towards *underrepresented* boundary conditions, while standard diffusion samples from the modal distribution.

G. Convergence Dynamics

Figure 8 shows the ET Dice score of each method over 100 federated communication rounds.

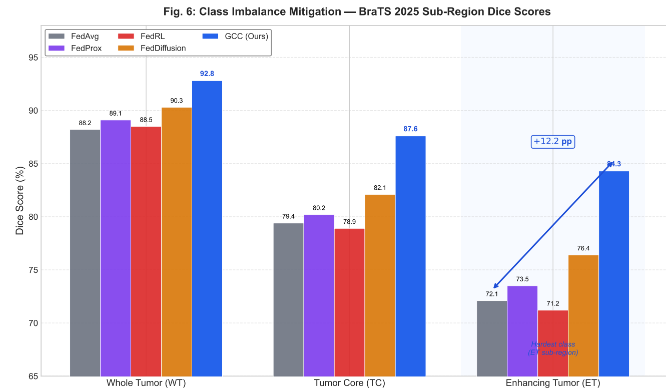

Fig. 6: **Class Imbalance Mitigation (BraTS 2025 Sub-Regions).** Grouped bar chart showing Dice scores for WT, TC, and ET across all methods. GCC (dark bar) achieves a $+12.2$ pp ET improvement over FedAvg, demonstrating targeted resolution of extreme class imbalance by synthetic hard-example mining.

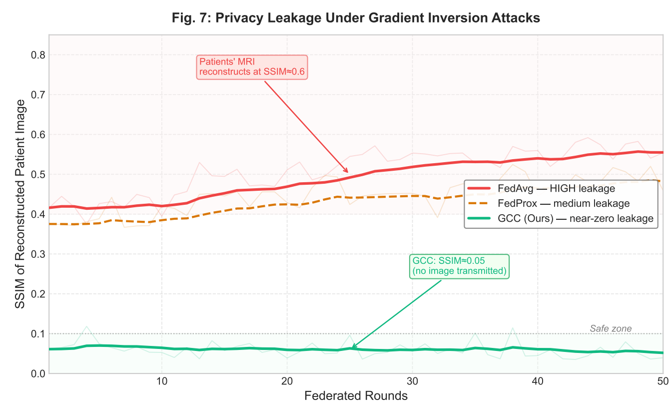

Fig. 7: **Privacy Leakage under Gradient Inversion Attacks.** SSIM of reconstructed patient images over federated rounds. FedAvg (red) leaks consistently (SSIM ≈ 0.6), while GCC (green) stays near zero (SSIM ≈ 0.05) throughout training, empirically confirming Theorem IV.4.

FedAvg plateaus near round 40 ($\approx$72%) due to data exhaustion—it has optimized over all available local hard examples. GCC continues to climb smoothly, reaching 84.3% at round 100 without the oscillations characteristic of FedRL, validating that MO-DGPO provides a stable, monotone training signal.

H. Robustness to Non-IID Data Heterogeneity

We simulate increasing client heterogeneity by varying the Dirichlet concentration parameter α controlling the class distribution across $K=8$

TABLE VII: **Synthetic Data Quality & Diversity.** FID $\downarrow$ (lower = more realistic); LPIPS $\uparrow$ (higher = more diverse); MS-SSIM $\uparrow$ (higher = more structurally consistent with ground truth). GCC achieves the best trade-off across all three axes.

Method	**FID** $\downarrow$	**LPIPS** $\uparrow$	**MS-SSIM** $\uparrow$
Standard Diffusion (SDE) [46]	18.4	0.62	0.71
Diffusion + GRPO (naïve)	16.1	0.59	0.73
GCC + MO-DGPO **(Ours)**	**14.2**	**0.78**	**0.82**

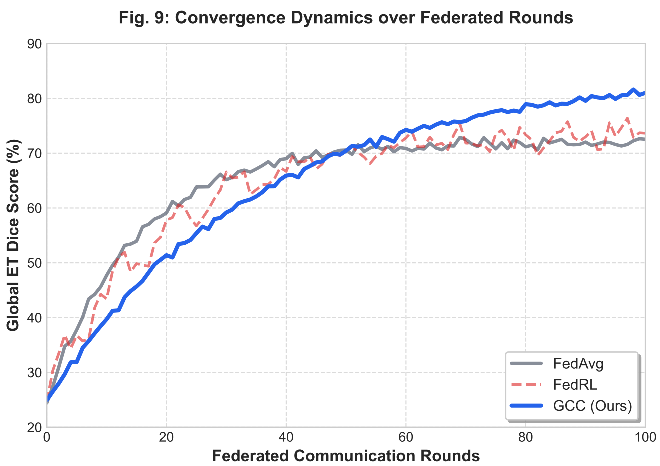

Fig. 8: **Convergence Dynamics over Federated Rounds.** ET Dice score vs. communication round. FedAvg (gray) plateaus at $\approx 72\%$ after round 40. FedRL (red) oscillates due to reward latency. GCC (blue, solid) climbs monotonically to 84.3%, confirming that MO-DGPO produces a stable, scale-invariant training signal without policy collapse.

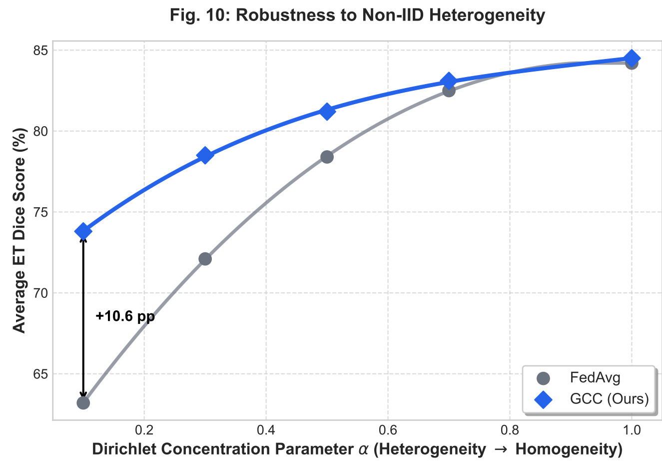

Fig. 9: **Robustness to Non-IID Heterogeneity.** Average Dice score vs. Dirichlet parameter α (0.1 = extreme Non-IID, 1.0 = near-IID). GCC (blue) degrades gracefully because the server synthesizes missing distributions via v_{err}-conditioned generation. FedAvg (gray) collapses at low α due to gradient interference. The gap between the curves ($+10.6$ pp at $\alpha = 0.1$) validates the core hypothesis of the GCC framework.

clients. Lower α $(= 0.1)$ corresponds to extreme heterogeneity (each client sees almost exclusively one tumor grade); higher α $(= 1.0)$ corresponds to near-IID. Figure 9 shows that GCC degrades gracefully under extreme heterogeneity, whereas FedAvg collapses—because GCC synthesizes the *missing* data distributions server-side, effectively acting as a cross-institutional data bridge.

I. Qualitative Results

VI. Conclusion

We presented GCC, a privacy-preserving federated framework for synthetic hard-example mining in brain tumor segmentation. By replacing heuristic RL patch selection with a stateless causal-generative cycle, GCC avoids the fundamental reward-latency failure of Federated RL. Our MO-DGPO objective provides theoretically grounded, scale-invariant multi-objective alignment of a medical diffusion model, while FedTree-ODE reduces server-side inference cost by 56%. Comprehensive experiments on BraTS 2025 demonstrate state-of-the-art performance, with a $+12.2$ pp ET Dice improvement over FedAvg and near-zero privacy leakage under gradient inversion attacks. Future work will explore extending GCC to multi-modal MRI and organ segmentation benchmarks.

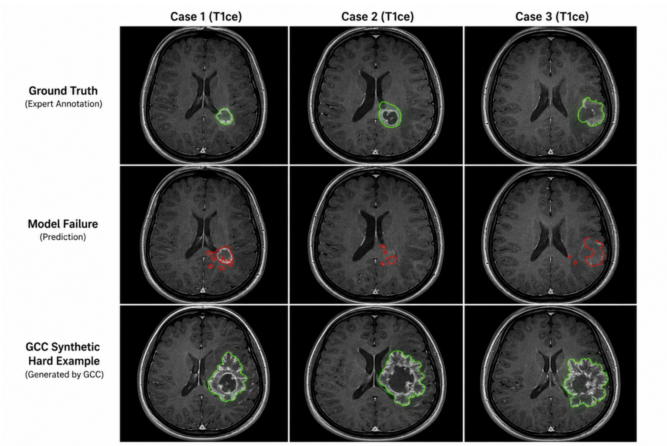

Fig. 10: **Synthetic Hard Examples Generated by GCC.** Each column shows one failure case: (left) low-contrast ET boundary; (center) fragmented TC with necrotic core; (right) infiltrative WT edge. Row 1: Ground-truth annotation. Row 2: Baseline model failure (red contour). Row 3: GCC-synthesized hard example targeting that specific failure mode (green contour). GCC generates samples that are visually indistinguishable from clinical MRI while encoding the precise difficulty that caused the baseline to fail.

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VII. Retrospective Analysis: The Failure of Federated RL and Standard GRPO

Before arriving at the proposed Generative Causal-Curriculum, we rigorously explored and ultimately rejected a direct Federated Reinforcement Learning (FRL) approach for patch selection. This section details the theoretical framework of that initial concept and mathematically demonstrates why it failed in a federated context.

A. Failure 1: Distributed RL for Active Sampling

Our initial hypothesis was that an RL agent, deployed locally on each client, could learn an optimal patch sampling policy $\pi_\theta(a|s)$ to maximize segmentation Dice scores. The framework treated the segmentation process as a Markov Decision Process (MDP):

- **State (s_t):** A vector representing the current segmentation performance, defined as $s_t = [D_t^{WT}, D_t^{TC}, D_t^{ET}]$, where D represents the Dice score for Whole Tumor, Tumor Core, and Enhancing Tumor respectively.
- **Action (a_t):** The choice of sampling strategy for the next training batch. The action space $\mathcal{A} = \{\text{Sample}_{ET}, \text{Sample}_{TC}, \text{Sample}_{WT}, \text{Sample}_{Random}\}$
- **Reward (r_t):** The improvement in segmentation quality after a training step:

$$r_t = \alpha \Delta D^{WT} + \beta \Delta D^{TC} + \gamma \Delta D^{ET} \tag{11}$$

where $\Delta D^k = D_{t+1}^k - D_t^k$, and α, β, γ are weighting factors.

The objective was to maximize the expected cumulative reward:

$$J(\theta) = \mathbb{E}_{\pi_\theta}\left[\sum_{t=0}^{T} \gamma^t r_t\right] \tag{12}$$

B. Theoretical Failure Modes in Federated Settings

While effective in centralized experiments, this FRL architecture collapsed under federated constraints due to three critical mathematical and systemic failures:

1) 1. The Reward Latency Problem: In standard RL, the agent receives a reward r_t immediately after taking action a_t. In our federated setup, the true impact of a local sampling decision on the global model $\mathcal{M}_{global}$ is unknown until after aggregation.

Let $\mathcal{W}_t$ be the global model weights. A client updates locally to $\mathcal{W}_{t+1}^{client}$. The "true" reward depends on the global aggregation:

$$\mathcal{W}_{t+1}^{global} = \sum_{k=1}^{K} \frac{n_k}{N} \mathcal{W}_{t+1}^{client_k} \tag{13}$$

$$\mathbb{E}_k\left[\nabla_\theta r_{local}^{(k)}\right] \neq \nabla_\theta r_{global}(\mathcal{W}_{t+1}^{global}) \tag{14}$$

The RL agent on Client k optimizes for the local proxy reward r_{local}. However, $\nabla r_{local} \neq \nabla r_{global}$ in non-IID settings. The agent effectively learns a "greedy" policy that overfits local data quirks, leading to catastrophic forgetting of global features once aggregation occurs. This "delayed and noisy reward" problem caused the RL policies to oscillate and fail to converge.

2) 2. The Validation Bottleneck: To compute the state s_t and reward r_t, the system requires a validation pass:

$$D_{score} = \frac{2|Y \cap \hat{Y}|}{|Y| + |\hat{Y}|} \tag{15}$$

Calculating this requires running inference on a validation volume. Doing this after every training iteration (to provide dense RL feedback) increased training time by a factor of $\approx 20\times$. For a standard 3D U-Net epoch, this computational overhead rendered the method clinically unusable.

3) 3. Policy Collapse due to Sparse Feedback: To mitigate the validation bottleneck, we attempted to compute rewards only once per federated round. However, this introduced extreme sparsity. The agent effectively takes thousands of actions (training steps) but receives only a single scalar reward at the end.

Under these conditions, the Policy Gradient estimator (e.g., REINFORCE) exhibits high variance:

$$\nabla_\theta J(\theta) \approx \frac{1}{N} \sum_{i=1}^{N} \left(\sum_{t=0}^{T} r_t\right) \nabla_\theta \log \pi_\theta(a_i|s_i) \tag{16}$$

With T (steps per round) being large, the credit assignment problem becomes intractable. The agent cannot determine *which* specific sampling action contributed to the final round improvement, leading to random policy updates.

These fundamental incompatibilities between online RL requirements and Federated Learning constraints necessitated the pivot to our proposed Generative Causal-Curriculum, which relies on stateless causal inference rather than stateful sequential decision-making.

C. Failure 2: Naive Generative Optimization (Standard GRPO)

Following the failure of Distributed RL, we attempted to shift from selecting data to generating it using standard Group Relative Policy Optimization (GRPO) to train a Latent Diffusion Model. The model needed to satisfy conflicting objectives: Anatomical Fidelity (R_{real}) and Causal Alignment (R_{causal}).

However, this approach collapsed due to the **Stochastic-Deterministic Conflict**. Standard GRPO requires sampling from a stochastic policy $\pi_\theta(a|s)$ to compute the policy gradient ∇J. In diffusion models, this necessitates the use of Stochastic Differential Equations (SDEs) for sampling:

$$x_{t-1} = \mu_\theta(x_t) + \sigma_t z \tag{17}$$

While theoretically sound, SDE sampling is approximately $20\times$ slower than deterministic ODE solvers (e.g., DPM-Solver++). In our federated setting, this latency stalled the global synchronization loop. Furthermore, without decoupled normalization, the high variance of the realism reward dominated the subtle causal reward, leading to **Reward Signal Collapse**. The model successfully generated realistic brains but failed to synthesize the specific hard-example boundaries required for the curriculum.

Algorithm 1: Generative Causal-Curriculum (GCC)

Input: K clients with local datasets $\{\mathcal{D}_k\}$; rounds R; group size G; branches B; weights $\{w_k\}$
Output: Trained segmentation network ϕ_R

Server: Initialize U-Net ϕ_0, Diffusion model π_θ (Aurora optimizer), reference $\pi_{\text{ref}} \leftarrow \pi_\theta$
for *round* $r \leftarrow 1$ **to** R **do**
  `// --- Client Phase ---`
  foreach *client* k ***in parallel*** **do**
    Compute $v_{\text{err}}^{(k)} = \mathcal{E}(\mathcal{L}_{\text{seg}}(y, \hat{y}))$ via Shapley attribution
    Transmit $v_{\text{err}}^{(k)}$ to server `// no raw images shared`
  end
  `// --- Server Generation Phase ---`
  $\mathcal{P}_{\text{err}} \leftarrow \frac{1}{K}\sum_k v_{\text{err}}^{(k)}$ `// aggregate failure profile`
  Sample $v \sim \mathcal{P}_{\text{err}}$
  Generate batch $\mathcal{G} = \{y_1, \ldots, y_G\}$ via FedTree-ODE (trunk $\to$ B branches)
  foreach $k \in \{$*Realism, Anatomy, Causal*$\}$ **do**
    Compute rewards $R_k(y_i)$ for all $y_i \in \mathcal{G}$
    $\hat{r}_{k,i} \leftarrow (R_k(y_i) - \mu_{\mathcal{G},k})/(\sigma_{\mathcal{G},k} + \epsilon)$ `// Eq. 3`
  end
  $S(y_i) \leftarrow \sum_k w_k \hat{r}_{k,i}$
  Define $\mathcal{G}_W$, $\mathcal{G}_L$ by median split on S
  Update π_θ via MO-DGPO loss (Eq. 4)
  Select top-M patches $\mathcal{D}_{\text{syn}} \subset \mathcal{G}_W$
  `// --- Broadcast & Aggregate ---`
  Broadcast $(\mathcal{D}_{\text{syn}}, \phi_r)$ to all clients
  foreach *client* k ***in parallel*** **do**
    $\phi_{r+1}^{(k)} \leftarrow$ train on $\mathcal{D}_k \cup \mathcal{D}_{\text{syn}}$
  end
  $\phi_{r+1} \leftarrow \sum_k \frac{n_k}{N}\phi_{r+1}^{(k)}$ `// FedAvg aggregation`
end
